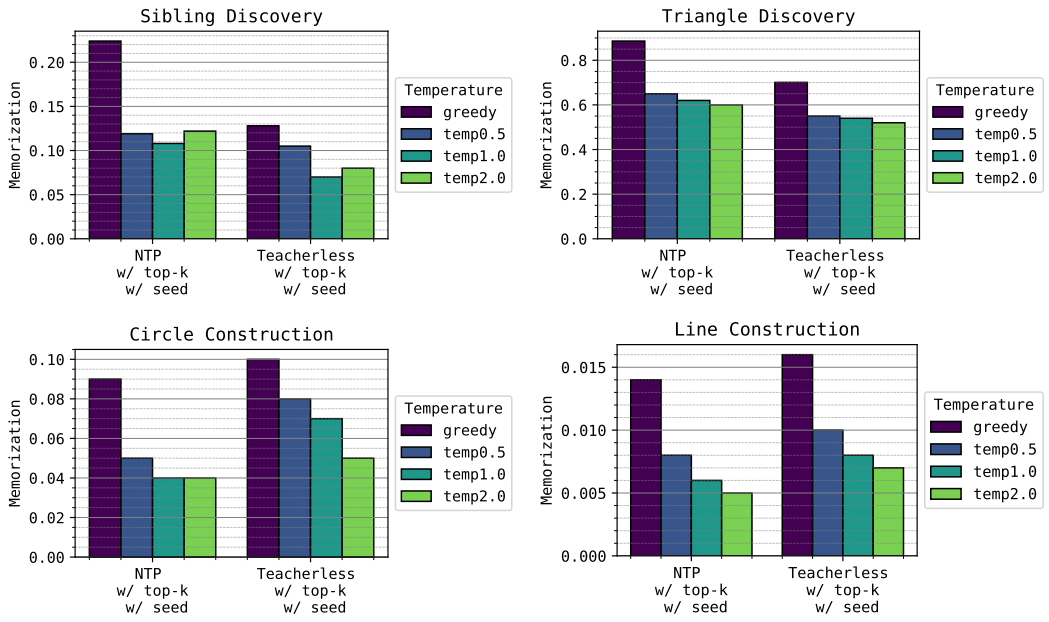

Figure 25. **Memorization scores for `NTP` vs teacherless `MTP` training with seed-conditioning + top-K sampling where $K = 50$.**

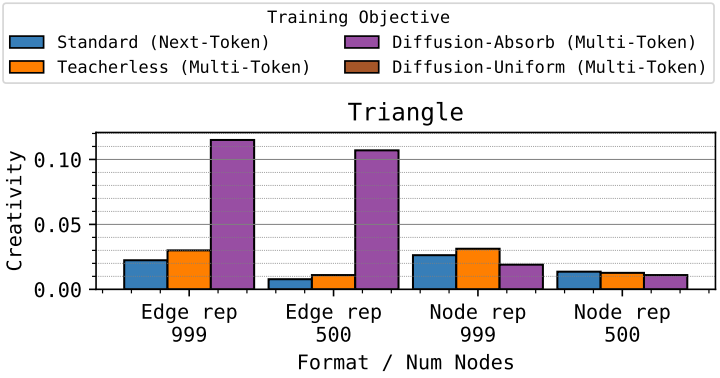

Figure 26. **Sensitivity to formatting of the sequence in `Triangle Discovery`:** We find that all our small models perform equally poorly with a node-wise representation of the input sequence, whereas there was a stark difference in performance with the edge-wise representation. We use greedy decoding in this plot.

G. Additional experiments with medium-sized Transformer and SEDD

We replicate our `SEDD (90M)` and `GPT-2 (86M)` experiments on a larger model size (∼400M parameters). In Fig 29, we see similar trends to the smaller model sizes (Fig 4).

H. Decomposing creativity

Through following experiments on the `GPT-2 (86M)` model in the `Sibling Discovery` task, we try to understand the dynamics between two important quantities that affect algorithmic creativity: diversity/duplication and originality/memorization.

H.1. Diversity score

Equation (1) defines our algorithmic creativity by rewarding samples that are both unique and novel. A higher score can be achieved either by enhancing diversity or by reducing memorization. In the following section, we examine this

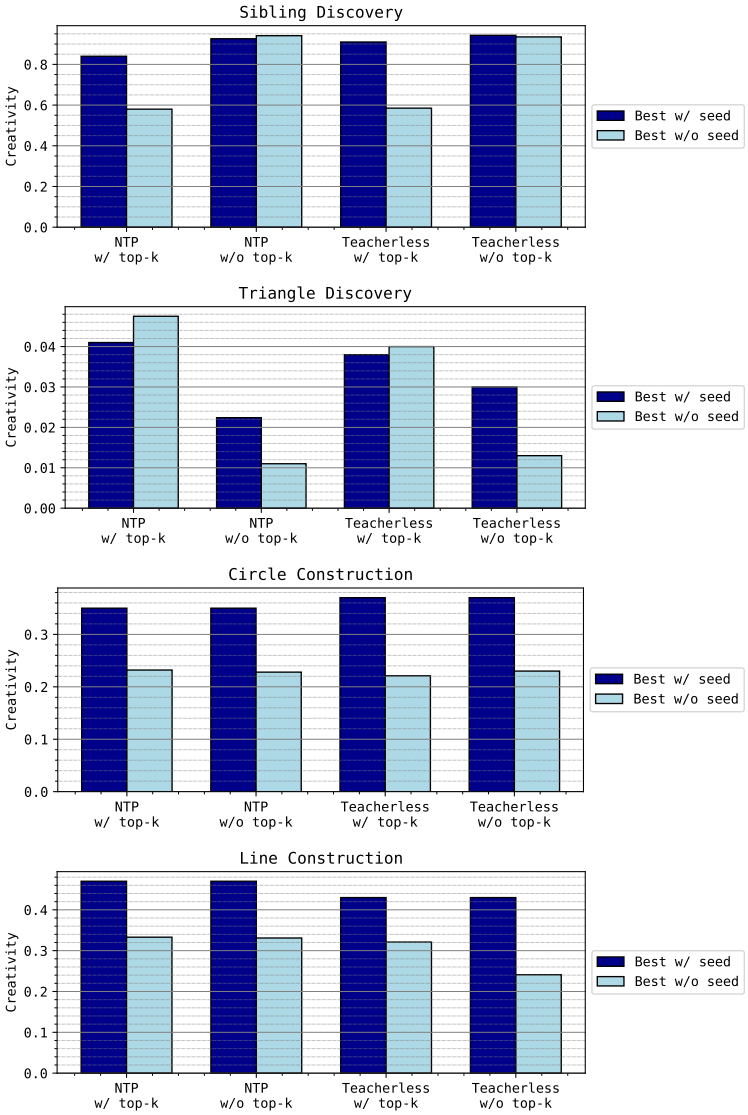

Figure 27. **Seed conditioning improves creativity.** "Best with seed" and "best without seed" stand for the best creativity score in each setting after we tune the sampling temperature from $\{0, 0.5, 1, 2\}$.

decomposition using the `Sibling Discovery` task. Formally, we define the diversity score as:

$$\hat{\mathtt{dv}}_N(T) = \frac{\mathtt{uniq}(\{\boldsymbol{s} \in T | \mathtt{coh}(\boldsymbol{s})\})}{|T|}. \tag{4}$$

We first demonstrate that creativity and diversity are not necessarily correlated, and next, that `MTP` particularly improves creativity (while achieving lower diversity than `NTP`). To show this, we report the algorithmic creativity and diversity scores along training in Fig 30. We see that for `NTP`, the diversity score keeps increasing and stays high, while algorithmic creativity increases in the first 10k steps and starts to decrease. For teacherless training, both scores increase throughout training. While the diversity of `MTP` is surprisingly lower than `NTP` throughout training, the creativity of `MTP` surpasses `NTP` at 20k steps.

H.2. Decomposing algorithmic creativity as diversity and memorization

Better creativity can be achieved either by enhancing diversity or by reducing memorization – we try to disentangle these factors in this section. In Fig 31, we plot the algorithmic creativity, diversity, and memorization scores at the checkpoint

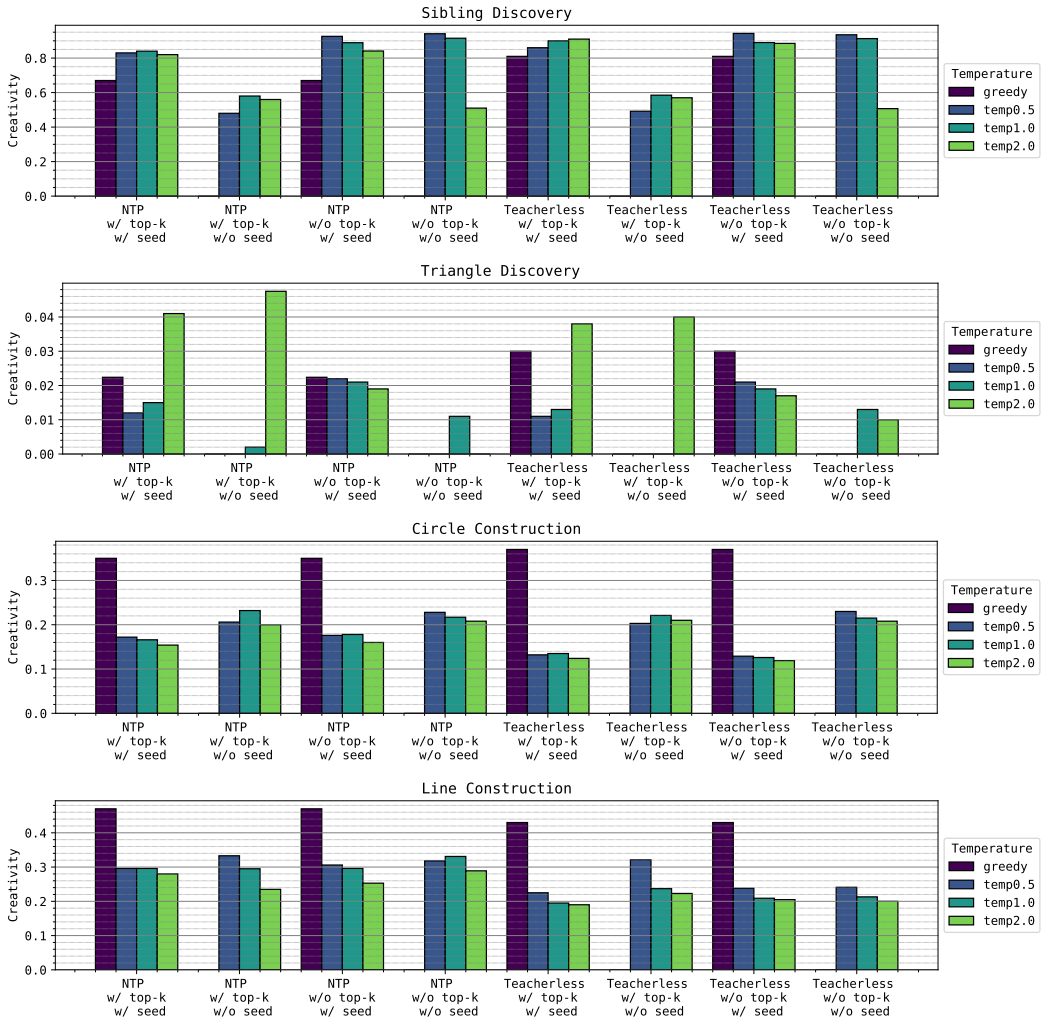

Figure 28. **Ablations on `GPT-2 (86M)` for (1) `NTP` vs teacherless `MTP` training, (2) effect of seed conditioning, (3) effect of top-K, and (4) effect of temperature sampling**

of best algorithmic creativity. We see that seed-conditioning contributes to higher diversity but does little to bring down memorization; however, teacherless `MTP` training contributes to higher diversity and also to reducing memorization. In Fig 30, we see that the best creativity and best diversity are not achieved at the same checkpoint.

H.3. Data scaling for algorithmic creativity

How does algorithmic creativity change as we increase the amount of training data? Intuitively, more training data helps the model learn the true distribution, but also makes it harder to generate unseen samples (since the uncovered space becomes rarer). To understand this, we plot how models perform relative to a *theoretically expected* maximum algorithmic creativity. This is computed by assuming an oracle that samples a generated set T (in Eq. (1)) uniformly with replacement from the true underlying distribution, and then computing algorithmic creativity Eq. (1. In Fig 32, we see that as we increase the training data (for a fixed underlying graph), the theoretically expected creativity decreases as expected, while the theoretically expected diversity stays the same (since this quantity does not care about being original with respect to the training set). Interestingly, as training data increases, `MTP` narrows the gap between `NTP` and the theoretically expected creativity and almost achieves the theoretically expected performance in the high data regime.